

Problem-Solution Fit

Date	30 june 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit

1. CUSTOMER SEGMENT(S) [CS] - Bank credit analysts & loan officers - Compliance officers & new applicants	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Very limited time per application - Legacy systems lack ML tools; data often incomplete	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual review: accurate but slow & costly Rule-based / single-model tools: rigid, no comparison
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Thousands of applications daily; manual screening can't keep up - Frequent rejections & human error in repetitive checks	9. PROBLEM ROOT / CAUSE [RC] - No automated, data-driven decision system - Manual rule-of-thumb, no standardized feature engineering	7. BEHAVIOR + ITS INTENSITY [BE] - Analysts cross-check data manually (High, daily) - Officers batch-review risky applicants (Medium-high)
3. TRIGGERS TO ACT [TR] - Surge in daily application volume - Audits flagging high-risk applicants	10. YOUR SOLUTION [SL] - ML model predicts approval/rejection instantly from data - Compares LR, Random Forest, XGBoost & Decision Tree - Deployed via Flask + IBM Watson ML (cloud)	8. CHANNELS OF BEHAVIOR [CH] ONLINE: Flask web app + IBM Watson ML cloud OFFLINE: In-branch analyst review
4. EMOTIONS BEFORE / AFTER [EM] Before: Frustration from slow, error-prone decisions After: Confidence from fast, accurate approvals		